AI5707 – Pattern Recognition and Machine Learning

Assignment 7: Gaussian Mixture Models for Handwritten Digit Classification

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1 Introduction

This assignment applies Gaussian Mixture Models (GMMs) to classify three handwritten MNIST digits assigned to Team 16: 3, 5, and 9. The task involves preparing the image data, reducing its dimensionality, fitting a class-specific GMM for each digit, selecting the number of Gaussian components using a development set, and evaluating the final classifier on an independent test set.

Unlike a single Gaussian distribution, a GMM can represent a class using several Gaussian components. This is useful for handwritten digits because one digit may have several writing styles, such as differences in slant, stroke thickness, curvature, or loop shape.

The probability density of a GMM with K components is

$$p(\mathbf{x}) = \sum_{i=1}^{K} \pi_i \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), \qquad \sum_{i=1}^{K} \pi_i = 1.$$

Here, π_i is the mixing weight of component i, $\boldsymbol{\mu}_i$ is its mean vector, and $\boldsymbol{\Sigma}_i$ is its covariance matrix. The model combines the component distributions to estimate how likely a feature vector is. During classification, the image is assigned to the digit model that gives the highest class-conditional log-likelihood.

2 Dataset and Pre-processing

The supplied TEAM_16 dataset contains separate training, development, and test splits. Each split has folders for digits 3, 5, and 9. Every image is a 28×28 grayscale image, giving 784 pixel features. The image is converted to grayscale, flattened into a one-dimensional vector, and normalized by dividing each pixel intensity by 255 so that values lie between 0 and 1.

The training split is used to learn the PCA transformation and fit the GMMs. The development split is used only to compare candidate component counts. The test split is held aside until the final evaluation so that it provides an independent estimate of model performance.

Table 1: Number of images in each dataset split.

Split	Digit 3	Digit 5	Digit 9	Total
Training	4886	4368	4776	14030
Development	1245	1053	1173	3471
Test	1010	892	1009	2911

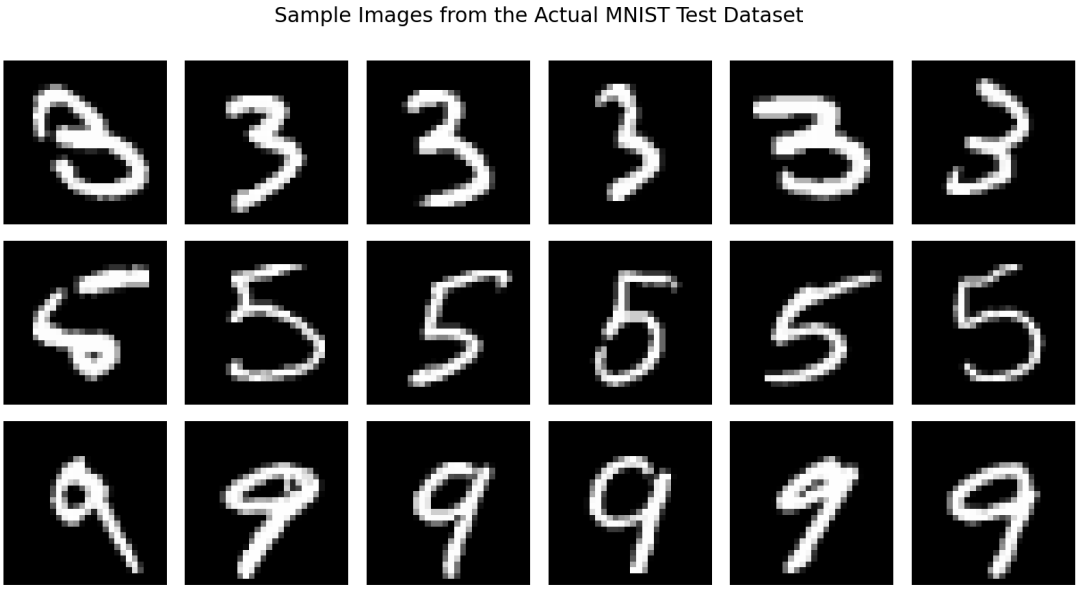


Figure 1: Sample images from the actual test dataset for digits 3, 5, and 9.

3 Dimensionality Reduction using PCA

Each image initially has 784 pixel features. Fitting GMMs directly in this high-dimensional space can be computationally expensive and can make covariance estimation more difficult. Principal Component Analysis (PCA) maps the image vectors into a lower-dimensional representation that preserves directions with high variance.

The notebook fits PCA using only the training images and retains 40 principal components. The transformation is then applied unchanged to development and test images. This avoids using information from the held-out splits when learning the transformation.

The transformation can be written as

$$\mathbf{z} = \mathbf{W}^{\mathsf{T}}(\mathbf{x} - \boldsymbol{\mu}),$$

where $\mathbf{x}$ is the original pixel vector, $\boldsymbol{\mu}$ is the training-set mean, and $\mathbf{W}$ contains the selected principal directions. In this run, PCA reduced the features from 784 to 40 and retained 79.73% of the variance.

4 GMM Training and Component Selection

A separate diagonal-covariance GMM is trained for each digit. Diagonal covariance models the variance of each PCA feature independently within a component and is more computationally manageable than estimating a full covariance matrix in the reduced feature space.

The number of components K controls how many Gaussian distributions are combined to represent each digit. A small K may not capture the variety of handwriting, while a larger K can model more complex patterns but may require more computation and can overfit. The notebook tests $K = 1, 2, 4, 6, 8$.

Table 2: Development-set accuracy for candidate numbers of GMM components.

Components per digit (K)	Development accuracy
1	92.51%
2	94.30%
4	95.36%
6	95.30%
8	95.74%

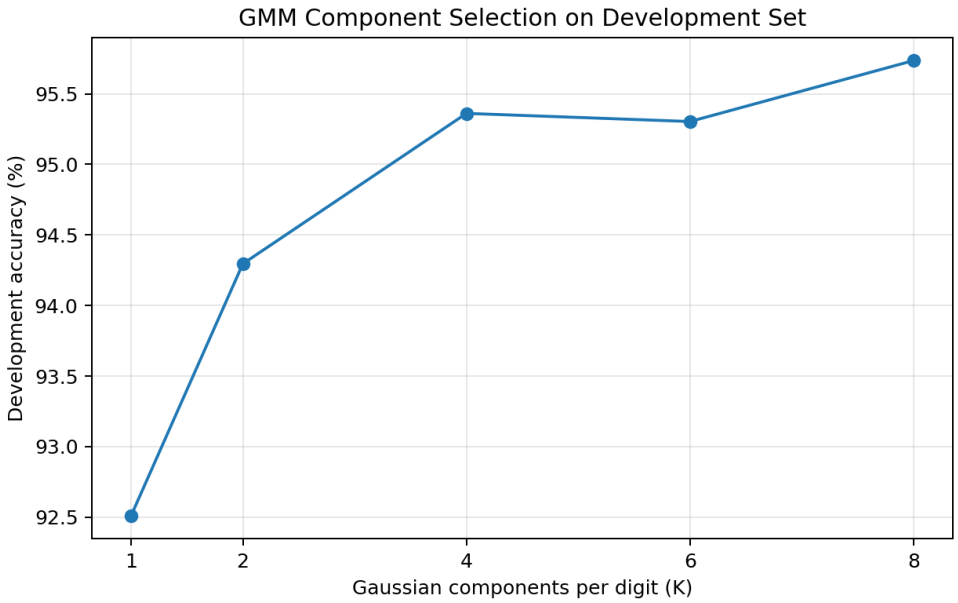


Figure 2: Development-set accuracy for the candidate numbers of GMM components.

The highest development accuracy was 95.74% at $K = 8$. Therefore, eight Gaussian components were selected for each digit. The test data was not used in this choice.

For a feature vector $\mathbf{z}$, the score for class c is $s_c(\mathbf{z}) = \log p(\mathbf{z} \mid C_c)$. The prediction rule is

$$\hat{y} = \arg \max_{c \in \{3,5,9\}} s_c(\mathbf{z}).$$

5 Test Classification and Evaluation

After selecting K, the trained class-wise GMMs were used to score every test image. Each test image was assigned to the digit with the largest log-likelihood. The test set contains 2,911 images, and 2,814 were classified correctly. The final test accuracy is 96.67%.

Accuracy is the proportion of all predictions that are correct:

$$\text{Accuracy} = \frac{\text{correctly classified images}}{\text{total test images}} = \frac{2814}{2911} = 96.67\%.$$

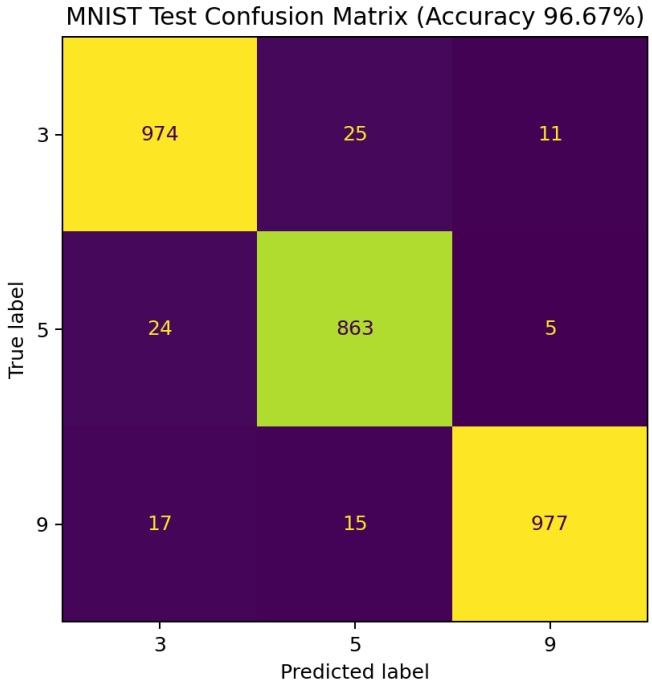


Figure 3: Confusion matrix for the held-out test set. Rows show true labels and columns show predicted labels in the order [3, 5, 9].

The confusion matrix (rows = true labels, columns = predicted labels in the order $[3, 5, 9]$) is

$$\begin{bmatrix} 974 & 25 & 11 \\ 24 & 863 & 5 \\ 17 & 15 & 977 \end{bmatrix}.$$

For example, 974 of the 1,010 true digit-3 images were correctly classified as 3; 25 were predicted as 5 and 11 were predicted as 9. Of the 892 true digit-5 images, 863 were classified correctly, 24 were predicted as 3, and 5 as 9. For digit 9, 977 of 1,009 images were classified correctly, while 17 were predicted as 3 and 15 as 5.

Precision measures how many predictions for a digit were correct. Recall measures how many actual images of that digit were found. The F1-score is the harmonic mean of precision and recall.

Table 3: Per-class test metrics.

Digit	Precision	Recall	F1-score	Support
3	95.96%	96.44%	96.20%	1010
5	95.57%	96.75%	96.16%	892
9	98.39%	96.83%	97.60%	1009
Macro average	96.64%	96.67%	96.65%	2911
Weighted average	96.68%	96.67%	96.67%	2911

The macro-average F1-score is 96.65% and the weighted-average F1-score is 96.67%. These values indicate that performance is consistently high across all three classes, with digit 9 having the strongest F1-score.

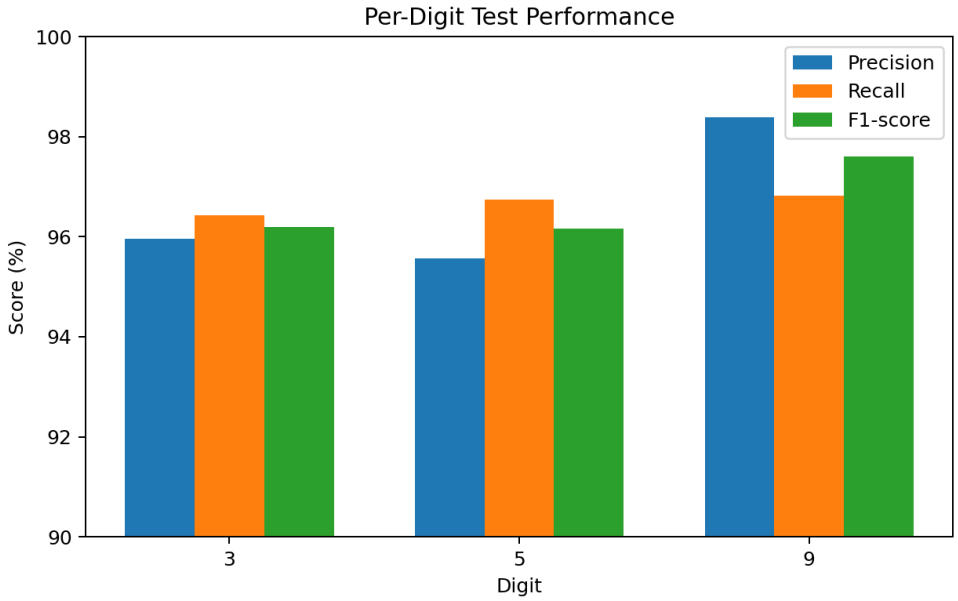


Figure 4: Precision, recall, and F1-score for each digit.

6 Explanation of Digit Confusions using GMM Boundaries

The largest off-diagonal entry in the confusion matrix is $3 \to 5$, with 25 images. The next largest is $5 \to 3$, with 24 images. There are also 17 cases of $9 \to 3$ and 15 cases of $9 \to 5$. This means that the main confusion occurs between handwritten digits 3 and 5.

A decision boundary between digits a and b is the set of feature vectors for which

$$\log p(\mathbf{z} \mid C_a) = \log p(\mathbf{z} \mid C_b).$$

On one side of the boundary, the digit-a model gives a higher likelihood; on the other side, the digit-b model gives a higher likelihood. An image is misclassified when a model for an incorrect digit gives a higher score than the model for its true digit.

Some handwritten 3s and 5s can have similar curves, stroke thickness, slant, or partially closed loops. These similarities may cause their PCA representations to fall in overlapping regions of feature space. PCA also compresses the original pixels, so some fine details that distinguish the digits may be weakened. The actual misclassified examples are shown below; they provide visual evidence for examining the model's errors.

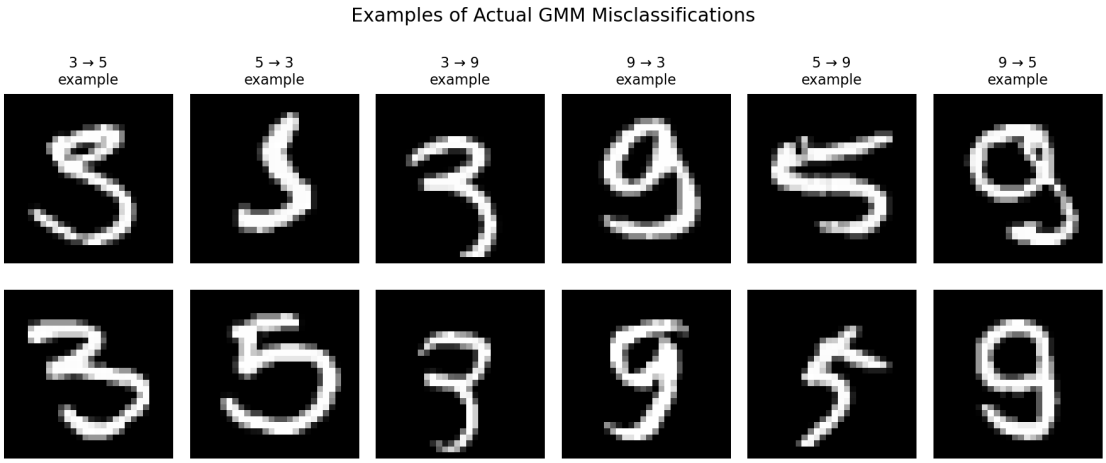


Figure 5: Examples of misclassified test images. Each title gives the true digit and the GMM-predicted digit.

7 Answers to Assignment Questions

Question 1 – Using the data generated in Assignment 6, build GMM models for each class using 80% of the data, perform classification, and create the confusion matrix.

Two synthetic classes are used for this part, with 1,500 observations in each class. The data are divided into an 80:20 split, giving 1,200 training and 300 test observations per class. A separate GMM is fitted to each class using its training observations, and each held-out sample is assigned to the class with the larger class-conditional likelihood. Figure 6 shows the training data used by the classifier. Figure 7 shows the learned GMM decision boundary and the held-out test samples.

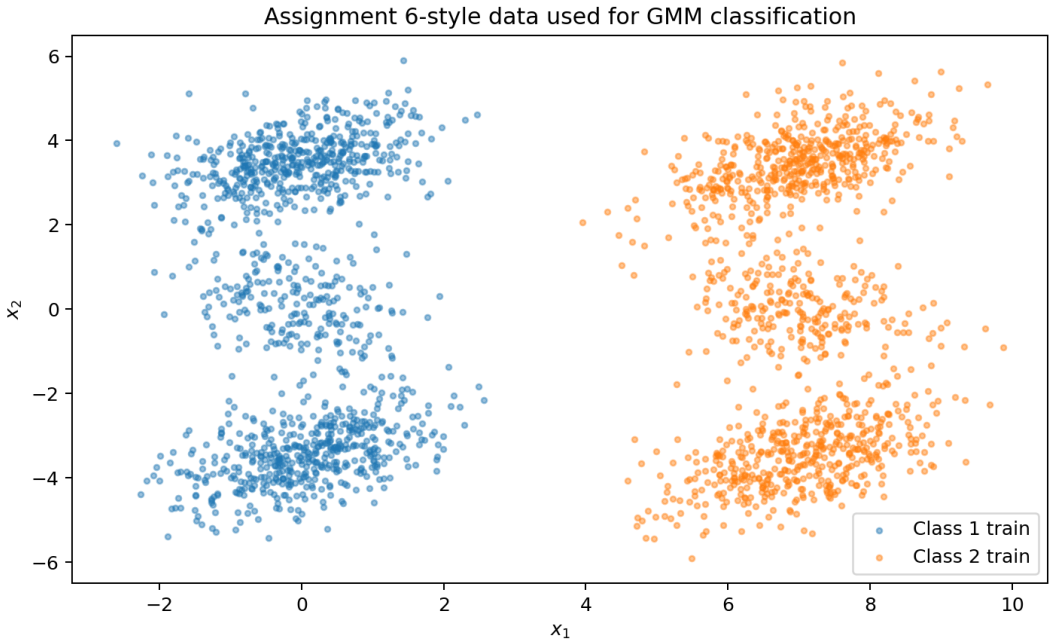


Figure 6: Question 1: Training observations for the two synthetic classes used to fit the GMM classifier.

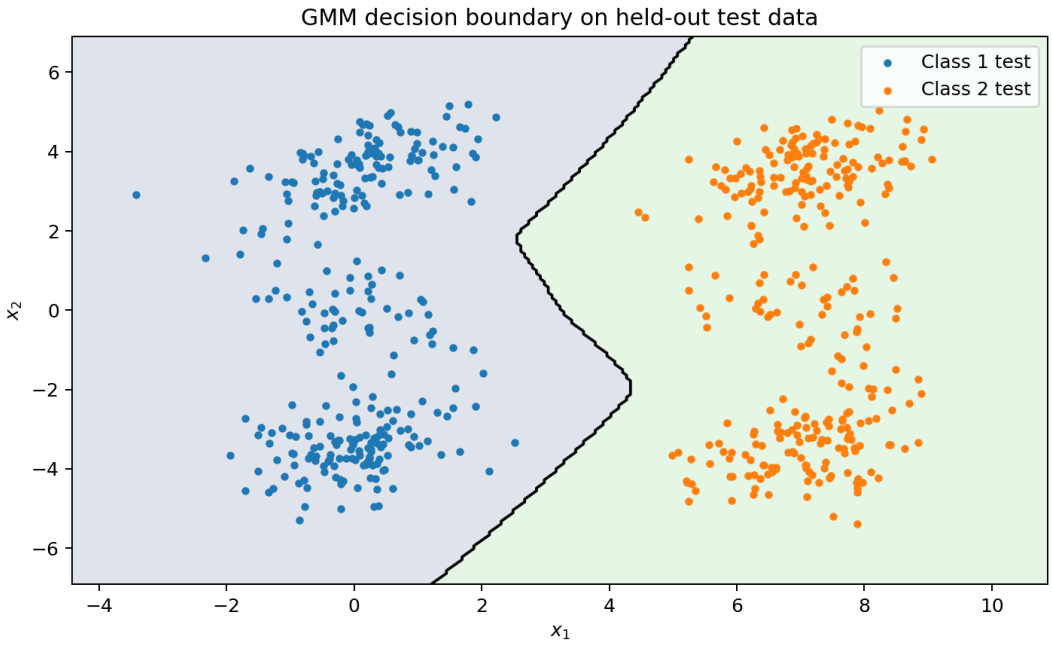


Figure 7: Question 1: GMM decision boundary on the held-out test data. The background indicates the class receiving the higher likelihood.

The resulting test confusion matrix is shown in Figure 8. All 600 test observations are classified correctly in this run, giving 100.00% test accuracy. The confusion matrix has 300 correct predictions for each class and zero off-diagonal errors.

$$\mathbf{C} = \begin{bmatrix} 300 & 0 \\ 0 & 300 \end{bmatrix}, \qquad \text{Accuracy} = \frac{300 + 300}{600} = 100.00\%.$$

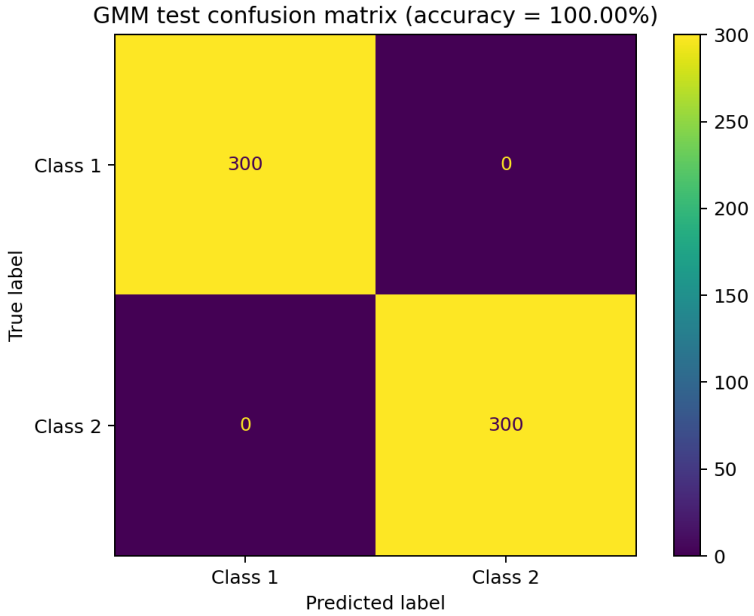


Figure 8: Question 1: Confusion matrix for the GMM classifier on the 600 held-out test observations.

Questions 2–4 concern the separate MNIST digit-classification experiment reported in the following sections.

Question 2 – Build a GMM model for each assigned digit. A separate diagonal-covariance GMM was fitted for each of digits 3, 5, and 9. PCA was learned using training images only, and the candidate values $K = 1, 2, 4, 6, 8$ were compared using development accuracy. $K = 8$ was selected because it gave the best development accuracy of 95.74%.

Question 3 – Classify the test data and record the results. The selected models achieved 96.67% accuracy on 2,911 test images, with 97 misclassifications. The confusion matrix, precision, recall, and F1-scores are presented in Section 5. The overall macro F1-score was 96.65%.

Question 4 – Explain the confused digits using the GMM models and class boundaries. The most frequent confusion was 3 predicted as 5 (25 cases), followed by 5 predicted as 3 (24 cases). This is consistent with overlapping class-conditional likelihoods near the boundary between the digit-3 and digit-5 models. The misclassified-image examples support a visual inspection of these errors.

8 Observations

1. Each MNIST image starts with 784 normalized pixel features.
2. PCA reduces the features to 40 dimensions and retains 79.73% of the variance.
3. The best development accuracy was 95.74%, achieved with 8 Gaussian components per digit.
4. The final test accuracy was 96.67% on 2,911 images.
5. Digit 9 achieved the highest F1-score at 97.60%.
6. Digits 3 and 5 were the most commonly confused pair, showing the importance of inspecting class-boundary errors.

9 Conclusion

This experiment trained class-wise Gaussian Mixture Models to classify handwritten MNIST digits 3, 5, and 9. PCA reduced the image representation from 784 to 40 features while retaining 79.73% of the variance. Development-set evaluation selected eight Gaussian components per digit. The final model achieved 96.67% accuracy on the held-out test set, with a macro F1-score of 96.65%. Most errors occurred between digits 3 and 5, which is consistent with overlapping likelihood distributions near their learned decision boundary. The generated confusion matrix, metric plot, component-selection curve, and misclassified examples provide the quantitative and visual evidence for the conclusions.